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Patent Public Search | Text View

United States Patent Application Publication

20250270077

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Fiegen; Joseph J. et al.

MOTOR MOVEMENT IN AN AIR HANDLER OF A CLIMATE CONTROL SYSTEM

Abstract

An embodiment of a method of moving a motor within an enclosure of an air handler of a climate control system includes (a) suspending the motor from a first lifting support and a second lifting support via lifting line, the first lifting support being laterally spaced from the second lifting support in the enclosure. In addition, the method includes (b) shortening the lifting line extending between the motor and the second lifting support. Further, the method includes (c) lengthening the lifting line extending between the motor and the first lifting support. Still further, the method includes (d) moving the motor laterally within the enclosure as a result of (b) and (c).

Inventors: Fiegen; Joseph J. (Onalaska, WI), Couch; Erick (Nicholasville, KY), Stewart; Joseph (Lexington, KY), Luebbehusen; Scott (Lexington, KY)

Applicant: Trane International Inc. (Davidson, NC)

Family ID: 1000007700363

Appl. No.: 18/584161

Filed: February 22, 2024

Publication Classification

Int. Cl.: B66D1/42 (20060101); B66D1/30 (20060101)

U.S. Cl.:

CPC B66D1/42 (20130101); B66D1/30 (20130101); B66D2700/0116 (20130101); B66D2700/0191 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable.

BACKGROUND

[0002] Climate control systems may cool or heat a conditioned space by use of a refrigerant circuit. Specifically, the climate control system may heat or cool the conditioned space by heating or cooling, respectively, an airflow with the refrigerant circuit (e.g., directly and/or via another thermally coupled fluid or circuit), and then circulating the heated or cooled airflow throughout the conditioned space.

[0003] In climate control systems for large buildings (e.g., such as office or industrial buildings, retail stores, etc.), the airflow maybe heated or cooled in an enclosed air handler and then supplied to the conditioned space from the air handler via ducting. Air handler may include enclosures that house one or more fan motors for driving airflow through the air handler during operations.

BRIEF SUMMARY

[0004] Some embodiments disclosed herein are directed to methods of moving a motor within an enclosure of an air handler of a climate control system. In some embodiments, a method includes (a) suspending the motor from a first lifting support and a second lifting support via lifting line, the first lifting support being laterally spaced from the second lifting support in the enclosure. In addition, the method includes (b) shortening the lifting line extending between the motor and the second lifting support. Further, the method includes (c) lengthening the lifting line extending between the motor and the first lifting support. Still further, the method includes (d) moving the motor laterally within the enclosure as a result of (b) and (c).

[0005] In some embodiments, a method includes (a) lifting the motor off of a motor base in the enclosure with a first lifting line that is connected to a first lifting support in the enclosure. In addition, the method includes (b) connecting the motor to a second lifting support with a second lifting line after (a), wherein the second lifting support is laterally closer to an access door of the enclosure than the first lifting support. Further, the method includes (c) lengthening the first lifting line between the motor and the first lifting support and (d) shortening the second lifting line between the motor and the second lifting support. Still further, the method includes (e) laterally moving the motor toward the access door as a result of (c) and (d).

[0006] Embodiments described herein comprise a combination of features and characteristics intended to address various shortcomings associated with certain prior devices, systems, and methods. The foregoing has outlined rather broadly the features and technical characteristics of the disclosed embodiments in order that the detailed description that follows may be better understood. The various characteristics and features described above, as well as others, will be readily apparent to those having ordinary skill in the art upon reading the following detailed description, and by referring to the accompanying drawings. It should be appreciated that this disclosure may be readily utilized as a basis for modifying or designing other structures for carrying out the same purposes as the disclosed embodiments. It should also be realized that such equivalent constructions do not depart from the spirit and scope of the principles disclosed herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] For a detailed description of various embodiments, reference will now be made to the accompanying drawings in which:

[0008] FIG. 1 is a schematic, overhead view of a climate control system illustrating an air handler according to at least some embodiments disclosed herein;

[0009] FIG. 2 is a schematic, overhead view of a climate control system illustrating another air handler according to at least some embodiments disclosed herein;

[0010] FIGS. 3A-3G are sequential, side views of a process of moving a motor within the enclosure of the air handler of FIG. 1 or 2 according to some embodiments disclosed herein;

[0011] FIGS. 4A-4C are sequential, side views of a process of moving a motor within the enclosure of the air handler of FIG. 1 or 2 according to some embodiments disclosed herein;

[0012] FIG. 5 is a side view of a come along that may be used to shorten or lengthen a lifting line in the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0013] FIG. 6 is a side view of a chain winch that may be used to shorten or lengthen a lifting line in the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0014] FIG. 7 is a schematic view of a motorized system for shortening or lengthening a lifting line in the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0015] FIG. 8 is a schematic view of another motorized system for shortening or lengthening a lifting line in the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0016] FIG. 9 is a side view of tapered wheels for paying lifting line in the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0017] FIGS. 10(a) and 10(b) are sequential views showing lifting line paid in and paid out from the tapered wheels of FIG. 9 during the processes of FIGS. 3A-3G or 4A-4C according to some embodiments disclosed herein;

[0018] FIG. 11 is a schematic view of a motorized system for moving a motor between two lifting supports that may be used in the processes of FIGS. 3A-3G or 4A-4C; and

[0019] FIGS. 12 and 13 are block diagrams of methods of moving a motor within an enclosure of an air handler according to some embodiments disclosed herein.

DETAILED DESCRIPTION

[0020] An air handler for a climate control system may include an enclosure that houses one or more fan motors therein. The enclosure may also form or define at least a portion of the airflow path of the air handler for ultimately supplying the conditioned (e.g., heated or cooled) airflow to the conditioned space during operations with the climate control system. As such, operating space within the enclosure may be limited so that movement of the fan motor(s) (e.g., into or out of the enclosure) may be difficult. Also, the fan motor(s) within the enclosure and an air handler may be heavy (e.g., reaching about 500 lbs. in some cases), which may further complicate movement of the fan motor(s) within the tight space of the enclosure. Moreover, the construction of complex lifting and maneuvering systems within an air handler enclosure may be unfeasible (e.g., due to cost, complexity).

[0021] Accordingly, embodiments disclosed herein include systems and methods for moving safely and efficiently moving a fan motor within the confined space of an enclosure of an air handler of a climate control system. In some embodiments, the systems and methods described herein may include controlled and coordinated lengthening and shortening of lengths of lifting lines connected to laterally spaced lifting supports within the enclosure so as to laterally traverse and maneuver the fan motor while minimizing uncontrolled movements (e.g., swinging). Thus, through use of the embodiments disclosed herein, a fan motor may be efficiently and safely moved about the limited space of an enclosure of climate control system air handler module with minimal additional infrastructure.

[0022] Referring now to FIG. 1, a climate control system **10** including an air handler **100** is shown according to some embodiments. The climate control system **10** may be configured to heat or cool a conditioned space **18** via an airflow **20** that is supplied to the conditioned space **18** during operations. The conditioned space **18** may include an interior of a relatively large, commercial building, such as an office building, retail store, industrial or manufacturing facility, sports facility, etc. Thus, the climate control system **10** may be configured to condition relatively large interior spaces/volumes and may, in some embodiments, be a chiller unit or other climate control system with a sufficiently large cooling/heating capacity for conditioning such larger spaces/volumes

during operations.

[0023] The air handler **100** generally includes an enclosure **110** having a plurality of vertically oriented walls **112**, as well as a ceiling and floor (not shown in FIG. **1**, but see ceiling **114** and floor **115** depicted in FIGS. **2A-2G**). The enclosure **110** may include an access door **116** that may allow personnel to enter the enclosure **110** (e.g., such as during maintenance and/or installation operations).

[0024] An inlet **120** may be positioned along one of the walls **112** that is configured to direct the airflow **20** into the enclosure **110** during operations. An outlet **122** is defined on another wall **112** (a different wall **112** from that associated with the inlet **120**) of the enclosure **110**. In some embodiments, the enclosure **110** may be generally rectangular in shape (e.g., such as a rectangular parallelepiped) such that the inlet **120** and outlet **122** are positioned on opposite (or opposing) parallel walls **112** of the enclosure **110** such as shown in FIG. **1**.

[0025] The inlet **120** may include one or more filters (not shown) for removing at least some particulates (e.g., dust, or other particulates) from the airflow **20** prior to entering enclosure **110**. The inlet **120** may receive the airflow **20** from suitable ducting **14**. In some embodiments, the airflow **20** may be at least partially recirculated from the conditioned space **18**. In addition, at least some of the airflow **20** may be sourced or provided from an ambient environment, such as the outdoor environment surrounding the building or structure forming or defining the conditioned space **18**.

[0026] A heat exchanger **130** (which may comprise one or a plurality of heat exchangers) may in fluid communication with the enclosure **110**. For instance, in some embodiments (such as is shown in FIG. **1**), the heat exchanger **130** may be positioned in the enclosure **110**, proximate or adjacent to the outlet **122**. In some embodiment, the heat exchanger **130** may be positioned in a separate enclosure (not shown) that is downstream of the enclosure **110**. In either case (e.g., whether heat exchanger **130** is positioned in the enclosure **110** or not), the heat exchanger **130** may be in fluid communication with the enclosure **110** so that the airflow **20** may eventually contact the heat exchanger **130** during operations.

[0027] The heat exchanger **130** may receive a flow of heat transfer fluid **13** from a refrigeration assembly **12** so that during operations, the heat exchanger **130** is configured to facilitate the transfer of heat between the heat transfer fluid **13** and the airflow **20** to thereby cool or heat the airflow **20**. For instance, in some embodiments, the climate control system **10** may be operated to cool the conditioned space **18**, so that the heat exchanger **130** may cool the airflow **20** via the heat transfer fluid **13**. Conversely, in some embodiments (e.g., such as when the climate control system **10** is configured as or includes a heat pump), the climate control system **10** may be operated to heat the conditioned space **18** so that the heat exchanger **130** may heat the airflow **20** via the heat transfer fluid **13**.

[0028] In some embodiments, the refrigeration assembly **12** may include or comprise a refrigerant circuit that is configured to circulate a refrigerant (e.g., hydrocarbons, fluorocarbons, hydrofluorocarbons, hydrochlorofluorocarbons, chlorofluorocarbons, ammonia, carbon dioxide, or some combination thereof) between a pair of heat exchangers (e.g., such as an evaporator and a condenser) to controllably change the phase of the refrigerant (e.g., between vapor and liquid) and thereby transfer heat between two mediums or spaces.

[0029] In some embodiments, the heat transfer fluid **13** may comprise the refrigerant so that the heat exchanger **130** may comprise a portion of the refrigerant circuit of refrigeration assembly **12** (in which case, the heat exchanger **130** may function either as an evaporator for the refrigerant when cooling the airflow **20** or as a condenser for the refrigerant when heating the airflow **20**). Alternatively, in some embodiments, the heat transfer fluid **13** may be separate from the refrigerant circuit of the refrigeration assembly **12**, and may instead comprise an additional, intermediary fluid circuit for transferring heat between the refrigerant of the refrigeration assembly **12** and the airflow **20** during operations. For instance (e.g., such as when climate control system **10** is configured as a

chiller), the heat transfer fluid **13** may comprise a water (or other aqueous solution) that exchanges heat with the refrigerant in the refrigeration assembly **12** and with the airflow **20** via the heat exchanger **130** in the air handler **100**. Specifically, in some embodiments, the heat transfer fluid **13** may comprise a chilled water circuit that is used to cool the airflow **20** via the heat exchanger **130** during operations.

[0030] A plurality of fan assemblies **150** are positioned in the enclosure **110** so as to drive the airflow **20** therethrough during operations. Each fan assembly **150** may include an air inlet **158**, an impeller **156**, and a motor **152**. The motor **152** may drive rotation of the impeller **156** via a shaft **154** so as to draw the airflow **20** into the air inlet **158** and discharge the airflow **20** toward the heat exchanger **130**. The motor **152** may be supported on a motor base **151**, which may comprise any suitable base, frame, stand, or other structure(s) that are configured to support the motor **152** in place during operations.

[0031] In some embodiments, the motors **152** may be heavy so that additional equipment may be needed to lift and maneuver the motors **152** during installation or maintenance thereof. In addition, the available space within the enclosure **110** may be limited so that use of traditional lifting equipment such as cranes and separate lifting frames and winches may not be practical or feasible. Accordingly, the air handler **100** may include an overhead lifting beam or support **160** that may be used to lift and move motors **152** (and/or other components of a fan assembly **150**) between their corresponding bases **151** and the access door **116**. As will be described in more detail herein, the lifting beam **160** may have a plurality of lifting supports **162** defined or mounted thereon that are configured to support one or more lifting lines for efficiently lifting and laterally moving the motors **152** within the enclosure **110** during operations. The lifting supports **162** may comprise any suitable brackets, attachments points, lifting eyes, posts, wheels (e.g., pulleys, gears, sprockets, etc.) or any other suitable structure or component that may be used to support the weight of one of the motors **152** via suitable lifting lines (e.g., lifting lines **170**, **172** described herein).

[0032] In the example embodiment shown in FIG. 1, the lifting beam **160** may be generally aligned with the access door **116** to facilitate maneuvering of the motors **152** toward and away from the access door **116**. However, as shown in FIG. 1, aligning the lifting beam **160** with the access door **116** may also cause the lifting beam **160** to be laterally offset from the motors **152**. Referring briefly to FIG. 2, in some embodiments, all or part of the lifting beam **160** may be positioned over (or substantially over) the motors **152** so as to help facilitate lifting and lowering the motors **152** vertically (or substantially vertically) relative to the corresponding motor base **151**. In some of these embodiments (as shown in FIG. 2), the lifting beam **160** may have a general L-shape or configuration including a first portion **161** extending over the motors **152** and a second portion **163** extending at about 90° from the first portion **161** to the access door. The first portion **161** may include a plurality of lifting supports **162** as previously described for the embodiment of lifting beam **160** shown in FIG. 1. In addition, the lifting beam **160** shown in FIG. 2 may include at least one lifting support on the second portion **163** that may generally be aligned with and/or proximate the access door **116**. As will be described in more detail herein, in some embodiments, the lifting beam **160** may be replaced with separate lifting supports (e.g., lifting supports **162**) that are individually connected to and supported by the structure of the enclosure **110** (e.g., walls **112**, ceiling **114**, floor **115**, etc.).

[0033] Referring now to FIGS. 3A-3G, a sequence of a process of lifting and moving one of the motors **152** within the enclosure **110** of the air handler **100** is shown according to some embodiments. As is described in more detail herein, the method shown in FIGS. 3A-3G may be accomplished by controllably paying in and paying out multiple lifting lines **170**, **172** that are connected to corresponding ones of the lifting supports **162** on lifting beam **160** and the motor **152** so as to laterally traverse the motor **152** through the enclosure **110**, between the door **116** and the corresponding motor base **151** with minimal equipment and additional infrastructure. The specific sequence shown in FIGS. 3A-3G illustrates a process of removing the motor **152** from the

enclosure **110** (and thus moving the motor **152** from the corresponding motor base **151** to or toward the door **116**); however, it should be appreciated that the process illustrated in FIGS. **3A-3G** (and described herein) may also be utilized to move the motor **152** within the enclosure **110** for other purposes (e.g., to install the motor **152**, move the motor **152** within the enclosure **110**, etc.).

[0034] Initially, as shown in FIGS. **3A** and **3B**, one of the motors **152** (which is identified in FIGS. **3A-3G** with reference number **152'**) is lifted off of the corresponding motor base **151** by a first lifting line **170** connected to a first of the lifting supports **162** (which is identified in FIGS. **3A-3G** with reference number **162A**). The first lifting line **170** may comprise any suitable elongate tether that is configured to be placed in tension to lift and suspend the motor **152'** during operations. For instance, the first lifting line **170** may comprise a cable, chain, rope, strap, chord, or some combination thereof. The first lifting line **170** may be connected to the motor **152'** via a lifting eye **153** (or any other suitable lifting point) defined on or connected to the motor **152'**.

[0035] In addition, the first lifting support **162A** may be generally positioned above (or substantially above) the motor **152'** so that the lifting line **170** may lift the motor **152'** vertically (or substantially vertically) off the base **151**. Referring briefly again to FIG. **2**, as previously described, in some embodiments, the lifting beam **160** or individual lifting supports **162** may be at least partially positioned vertically above the motors **152** to facilitate substantially vertical lifting or lowering relative to the motor bases **151**. However, referring briefly again to FIG. **1**, in some embodiments the lifting beam **160** or individual lifting support **162** may be spaced from the motors so that that lifting or lowering of a motor **152** relative to the corresponding motor base **151** may include some lateral deviation via the lifting supports

[0036] As shown in the sequence from FIGS. **3A** to **3B**, the motor **152'** may be lifted off the motor base **151** by reducing a length of the first lifting line **170** extending between the first lifting support **162A** and the motor **152'**. Any suitable method of reducing the length of the first lifting line **170** may be used (e.g., a winch, pulley, come along, etc.), and further details of potential systems and methods are described in more detail herein. As a result, further details of the process of shortening or paying in the first lifting line **170** are omitted from this specific description for the sake of brevity.

[0037] As shown in FIGS. **3C** and **3D**, once the motor **152'** is suspended from the first lifting support **162A** via the first lifting line **170**, the motor **152'** may then be traversed laterally within the enclosure **110** by connecting a second lifting line **172** between the motor **152'** and a second one of the lifting supports **162** (which is identified in FIGS. **3A-3G** with reference number **162B**). The second lifting support **162B** may be laterally spaced from the motor the first lifting support **162A** along the support beam **160**. As a result, as shown in FIG. **3C**, when the second lifting line **172** is initially connected to the motor **152'** and second lifting support **162B**, the second lifting support **162B** may also be laterally spaced from the motor **152'**. Because the process shown in FIGS. **3A-3G** is configured to move the motor **152'** from the motor base **151** to the access door **116**, the second lifting support **162B** may be laterally closer to the access door **116** than the first lifting support **162A**.

[0038] As shown in the sequence from FIG. **3C** to FIG. **3D**, once the motor **152'** is connected to each of the lifting supports **162A**, **162B** via the lifting lines **170**, **172**, respectively, lifting lines **170**, **172** may be controllably lengthened and shortened, respectively, so as to laterally shift the motor **152'** away from the first lifting support **162A** and toward the second lifting support **162B**. Specifically, the first lifting line **170** may be lengthened or extended between the first lifting support **162A** and the motor **152'** while the second lifting line **172** may be shortened between the second lifting support **162B** and the motor **152'** so that the motor **152'** may be moved laterally within the enclosure **110**. The lateral movement of the motor **152'** may continue until the motor **152'** is vertically aligned under the second lifting support **162B** and thus has moved laterally between the lifting supports **162A**, **162B**.

[0039] Referring now to FIG. **3E**, once the motor **152'** is vertically aligned under the second lifting

support **162B**, the first lifting line **170** may be disconnected from the first lifting support **162A**. Thereafter, the first lifting line **170** (or another lifting line) maybe connected to the motor **152'** and a third of the lifting supports **162** (which is identified in FIGS. **3A-3G** with reference numeral **162C**). The third lifting support **162C** may be laterally spaced from both the second lifting support **162B** and the motor **152'**, and once again may be laterally closer to the access door **116** than the second lifting support **162B**. Thus, as shown in the sequence from FIG. **3E** to FIG. **3F** and then from FIG. **3F** to FIG. **3G**, after the motor **152'** is connected to the lifting supports **162B**, **162C** via the lifting lines **172**, **170**, respectively, the motor **152'** may once again be moved laterally by controllably lengthening and shortening the lifting lines **172**, **170** in a similar manner to that described above when moving the motor **152'** between the lifting supports **162A**, **162B**. Specifically, the second lifting line **172** may be lengthened between the second lifting support **162B** and the motor **152'** while the first lifting line **170** may be shortened between the third lifting support **162C** and the motor **152'** so that the motor **152'** translates laterally between the lifting supports **162B**, **162C**.

[0040] The above-described process may be repeated a number of times using different pairs of the lifting supports **162** so as to ultimately traverse the motor **152'** across the enclosure **110** to the access door **116**, from which personnel may maneuver the motor **152'** utilizing larger lifting components (e.g., independent cranes, winches, etc.) that may be positioned outside the enclosure **110**.

[0041] Thus, by selectively and controllably lengthening and shortening lifting lines (e.g., lifting lines **170**, **172**) from laterally spaced lifting supports **162**, the motor **152'** may be laterally traversed across the enclosure **110** without the use of cumbersome cranes or other large lifting components such as overhead trolleys, which may not be usable in the confined space of the enclosure **110** and which may require more complex and/or expensive design and fabrication. It should be appreciated that the number of stages may be altered so that fewer or more lifting points and steps may be utilized to fully laterally translate the suspended motor **152'** across the enclosure. For instance, as shown in FIGS. **4A-4C**, an alternative lifting process is shown in which motor **152'** is traversed across the entire enclosure **110** in a single stage by connecting to two lifting supports **162** along the lifting beam **160**.

[0042] Specifically, as shown in FIG. **4A**, the motor **152'** may initially be lifted by the first lifting line **170** connected to a first lifting support **162A** that is substantially vertically aligned with the motor **152'** as previously described. Thereafter, as shown in sequence FIG. **4B**, the second lifting line **172** is connected to the motor **152'** and another lifting support **162D** that is positioned at or proximate to the access door **116**. Thereafter, as shown in the sequence from FIG. **4B** to FIG. **4C**, the first lifting line **170** is lengthened between the motor **152'** and the first lifting support **162A** and the second lifting line **172** is shortened between the motor **152'** and the lifting support **152D** so as to move the motor **152'** laterally across the enclosure **110**, from above the motor base **151** to the access door **116** in a single step. Without being limited to this or any other theory, the choice of the number of stages or steps for moving the motor **152'** across the enclosure **110** may be based on a number of factors, including, for instance, the weight of the motor, the total distance of travel for the motor **152'** to the access door **116**, the length and/or tensile strength of the lines **170**, **172**, etc.

[0043] Referring briefly again to FIG. **2**, as previously described, in some embodiments, the lifting beam **160** may be substantially L-shaped, having a first portion **161** that is aligned with the plurality of motors **152**, and a second portion **163** that extends from the first portion **161** to (or proximate to) the access door **116**. When moving a motor **152** between the access door **116** and one of the motor bases **151**, the motor **152** may be moved, using the lifting lines **170**, **172** via the methods shown in FIGS. **3A-3G** and **4A-4C**, between a lifting support **162** positioned on the first portion **161** and the at least one lifting support **162** positioned on the second portion **163** in order to move the motor **152** toward or away from the access door **116** during operations.

[0044] As previously described, the lifting lines **170**, **172** used in the methods shown in FIGS. **3A-**

3G and 4A-4C may be lengthened or shortened using any suitable method or device in order to affect the lateral movement of the motor **152** across the enclosure **110**. For instance, the lifting lines **170**, **172** may be lengthened or shortened via a manual or motorized device.

[0045] For example, FIGS. **5** and **6**, show some embodiments of a manual winch that may be used to lengthen or shorten the lifting lines **170**, **172** during the above-described lifting methods. Specifically, FIG. **5** shows a manual come-along **180** that includes a ratcheted handle assembly **182** that may be manually reciprocated by a user to either shorten or lengthen the corresponding lifting line (shown as the first lifting line in FIG. **5** as an example). The ratcheted handle assembly **182** includes a hook or eye **184** that may be engaged with the corresponding lifting support **162** (which is shown configured as a post or projection extending laterally outward from the lifting beam **160**) during operations. The lifting line **170** (or lifting line **172**) may be configured as a metal (e.g., steel) cable when a come-along or other similar ratcheting winch is used to controllably shorten or lengthen the lifting line during operations. In some embodiments, the lifting line may comprise a ratchet strap that is similarly adjustable (lengthwise) by actuating a ratcheting handle in a similar fashion to that described for the come along **180**.

[0046] As another example, FIG. **6** shows a manual chain hoist **190** that includes a frame or body **192** that supports the lifting line (shown as the first lifting line **170** as an example). The lifting line **170** (or lifting line **172**) may be configured as a length of chain when a chain hoist **190** is used to controllably shorten or lengthen the lifting line **170** during operations. The body **192** includes a hook or eye **194** that may be engaged with the corresponding lifting support **162** (which again is shown configured as a post or projection extending laterally outward from the lifting beam **160**) during operations. An adjustment chain **196** may be coupled to the body **192**. During operations, a user may manually pull on one side/leg of the adjustment chain **196** to shorten the lifting line **170** and may pull on the other, opposite side/leg of the lifting chain **196** to lengthen the lifting line **170**.

[0047] In some embodiments, one or more of the lifting lines **170**, **172** may be lengthened or shortened using a motorized wheel in order to affect the lateral movement of the motor **152** across the enclosure **110**. For instance, FIGS. **7** and **8** show examples where a pair of lifting supports **162A**, **162B** are or include wheels **202A**, **202B** connected to the lifting beam **160** (not shown) or other suitable structure(s) in the enclosure **110**. The wheels **202A**, **202B** are configured to controllably pay out and pay in the first lifting line **170** and second lifting line **172**, respectively, in order to laterally move or shift the motor **152**. The wheels **202A**, **202B** may comprise pulleys, sprockets, gears, drums, or any other suitable wheel that may rotate to pay in or pay out a lifting line as described herein.

[0048] In the embodiment shown in FIG. **7**, each of the wheels **202A**, **202B** may be independently driven by a separate driver **204A**, **204B**, respectively. The drivers **204A**, **204B** may comprise any suitable motor such as, for instance, electric motors, hydraulic motors, pneumatic motors, combustion engines, etc. In order to laterally shift the motor **152** from position (I) to position (II) (e.g., to the left) as shown in FIG. **7**, the driver **204B** may rotate the wheel **202B** to pay in (and thereby shorten) the second lifting line **172**, and the driver **204A** may rotate the wheel **202A** to pay out (and thereby lengthen) the first lifting line **170**. Conversely, in order to laterally shift the motor **152** from position (II) to position (I) (e.g., to the right) as shown in FIG. **7**, the driver **204B** may rotate the wheel **202B** to pay out (and thereby lengthen) the second lifting line **172**, and the driver **204A** may rotate the wheel **202A** to pay in (and thereby shorten) the first lifting line **170**. When moving the motor **152** between the positions (I) and (II) in FIG. **7**, the lifting lines **170**, **172** may be paid in and out at the same time or at different times; however, in either case the coordinated pay in and pay out of the lines **172**, **170** causes the motor **152** to shift laterally between the positions (I) and (II).

[0049] In the embodiment shown in FIG. **8**, each of the wheels **202A**, **202B** may be driven by a common driver **206** via a transmission **208**. The driver **206** may comprise any suitable driver or motor, such as previously described herein for the drivers **204A**, **204B**. In addition, the

transmission **208** may comprise any suitable device or system for transferring the output (e.g., rotational output) from the driver **206** to the wheels **202A**, **202B**. For instance, the transmission **208** may comprise one or more drive belts, gears, chains, pulleys, etc.

[0050] The transmission **208** may be configured to convert and transfer the output from driver **206** to drive suitable rotations of the wheels **202A**, **202B** to laterally shift the motor **152** between the positions (I) and (II). For instance, the transmission **208** may be configured to convert the output of driver **206** to pay out the first lifting line **170** from the first wheel **202A** and also to pay in the second lifting line **172** from the second wheel **202B** and thereby shift the motor **152** from the first position (I) to the second position (II) (or to the left) as shown in FIG. **8**. Conversely, the transmission **208** may be configured to convert the output of driver **206** to pay out the second lifting line **172** from the second wheel **202B** and also to pay in the first lifting line **170** from the first wheel **202A** and thereby shift the motor **152** from the second position (II) to the first position (I) (or to the right) as shown in FIG. **8**.

[0051] For both example embodiments shown in FIGS. **7** and **8**, a controller **210** may be communicatively coupled (via any suitable wired and/or wireless connection) to the drivers **204A**, **204B** (FIG. **7**) or drivers **206** (FIG. **8**) so as to control the rotation of wheels **202A**, **202B** to laterally move the motor **152** as previously described. For instance, with respect to the example embodiment shown in FIG. **7**, the controller **210** may control the rates of rotation of the wheels **202A**, **202B** via the drivers **204A**, **204B**, respectively, so that the motor **152** is moved laterally between the positions (I) and (II) without substantially any or with minimal vertical deviation.

[0052] Specifically, the controller **210**, when moving the motor **152** from the first position (I) to the second position (II) (or to the left) as shown in FIG. **7**, the controller **210** may initially rotate the first wheel **202A** faster than the second wheel **202B** via the drivers **204A** and **204B**, respectively, so as to pay out the first lifting line **170** from the first wheel **202A** faster than the second lifting line **172** is paid into the second wheel **202B** until the motor **152** has reached a substantial mid-point between the positions (I) and (II). Thereafter, the controller **210** may rotate the second wheel **202B** faster than the first wheel **202A** so as to pay in the second lifting line **172** into the second wheel **202B** faster than the first lifting line **170** is paid out from the first wheel **202A** and thereby to continue moving the motor **152** toward the second position (II). During these adjustments by the controller **210**, the rotational rates of the wheels **202A**, **202B** may be constantly varied as the motor **152** is traversed laterally between the positions (I), (II). Specifically, when moving from the first position (I) to the second position (II), the second wheel **202B** may have a rotational speed that is gradually increased along a continuous rate (e.g., a linearly varying rate), while the first wheel **202A** may have a rotational speed that is gradually decreased along a continuous rate (e.g., another linearly varying rate). The varying rates of speed for the first wheel **202A** and second wheel **202B** may find a common point generally midway between the positions (I) and (II), when the lines **170**, **172** are generally equally paid out from the wheels **202A**, **202B**, respectively.

[0053] The above-described differences in the rotational speed of the wheels **202A**, **202B** may allow the motor **152** to move in a substantially laterally direction without little, no (or substantially no) vertical deviation. As a result, during operations the motor **152** may be substantially maintained in a single laterally oriented plane (or lateral plane) when moving between the positions (I) and (II).

[0054] In addition, with respect to the example embodiment shown in FIG. **8**, the controller **210** may control the rates of rotation of the wheels **202A**, **202B** to prevent or reduce vertical deviation of the motor **152** during movement between the positions (I) and (II) as previously described by controlling one or both of the driver **206** and transmission **208**. For instance, the controller **210** may alter a rotational speed of the driver **206** and/or may adjust an output of the transmission **208** (e.g., via a change to gearing ratios, gear/belt selection, etc.) so as to controllably vary the rotational speeds of the wheels **202A**, **202B** as previously described for the embodiment shown in FIG. **7**.

[0055] For both of the example embodiments of FIGS. **7** and **8**, the controller **210** may adjust the speeds of the wheels **202A**, **202B** via the drivers **204A**, **204B** by tracking the rotations of the

wheels **202A**, **202B** and based on the known dimensions of the system (e.g., the lateral distance between the wheels **202A**, **202B** and/or positions (I) and (II), the initial paid lengths of the lifting lines **170**, **172**, etc.). Thus, the controller **210** and/or the drivers **204A**, **204B** may comprise, include (or be communicatively coupled to), devices that are configured to track or detect the rotations or rotational positions of the wheels **202A**, **202B** (e.g., either directly or via the rotational positions of other components such as the drivers **204A**, **204B**, **206**, one or more components of transmission **208**, etc.). For instance, in some embodiments the drivers **204A**, **204B** shown in FIG. 7 (or similarly the driver **206** shown in FIG. 8) may comprise servo motors or stepper motors that are configured to provide an output to the controller **210** that is indicative of the rotational positions or number of rotations of the drivers **202A**, **202B**, which can then be used by the controller **210** to determine the position of the motor **152** at or between the positions (I) and (II) and thereby adjust the rotational speeds of the wheels **202A**, **202B** via drivers **204A**, **204B** accordingly in order to avoid or minimize vertical deviation of the motor **152** as previously described.

[0056] The controller **210** may comprise one or more computing devices, such as a computer, tablet, smartphone, server, circuit board, or other computing device(s) or system(s). Thus, controller **210** may include a processor **212** and a memory **214**.

[0057] The processor **212** may include any suitable processing device or a collection of processing devices. In some embodiments, the processor **212** may include a microcontroller, central processing unit (CPU), graphics processing unit (GPU), timing controller (TCON), scaler unit, or some combination thereof. During operations, the processor **212** executes machine-readable instructions (such as machine-readable instructions **216**) stored on memory **214**, thereby causing the processor **212** to perform some or all of the actions attributed herein to the controller **210**. In general, processor **212** fetches, decodes, and executes instructions (e.g., machine-readable instructions **216**). In addition, processor **212** may also perform other actions, such as, making determinations, detecting conditions or values, etc., and communicating signals. If processor **212** assists another component in performing a function, then processor **212** may be said to cause the component to perform the function.

[0058] The memory **214** may be any suitable device or collection of devices for storing digital information including data and machine-readable instructions (such as machine-readable instructions **216**). For instance, the memory **214** may include volatile storage (such as random-access memory (RAM)), non-volatile storage (e.g., flash storage, read-only memory (ROM), etc.), or combinations of both volatile and non-volatile storage. Data read or written by the processor **212** when executing machine-readable instructions **216** can also be stored on memory **214**. Memory **214** may include “non-transitory machine-readable medium,” where the term “non-transitory” does not include or encompass transitory propagating signals.

[0059] The processor **212** may include one processing device or a plurality of processing devices that are communicatively coupled to one another (and potentially distributed in a variety of locations). Likewise, the memory **214** may include one memory device or a plurality of memory devices that are communicatively coupled to one another (and potentially distributed in a variety of locations). Thus, the controller **210** may comprise a plurality of individual “controllers” (e.g., such individual controllers that are coupled to the drivers **204A**, **204B**, transmission **208**, driver **206**, etc.).

[0060] In some embodiments, the varying rates of speeds of pay in and pay out of the lifting lines **170**, **172** may be accomplished via the mechanical structure of the wheels **202A**, **202B** (or other components such as the transmission **208** for the example embodiment shown in FIG. 8). For instance, as shown in FIG. 9, the wheels **202A**, **202B** may be conical in shape so as to affect different pay in and pay out rates of the lifting lines **170**, **172** during the lateral movement of the motor **152** and for a constant rotational speed of the wheels **202A**, **202B**. Specifically, in the embodiment shown in FIG. 9, each of the wheels **202A**, **202B** includes a central or longitudinal axis **225** (or axis of rotation), a first or inner end **220** and a second or outer end **221** opposite the

inner end **220**. The inner end **220** may be closest or most proximate to the corresponding driving component of the wheel **202A**, **202B** (e.g., such as a corresponding one of the drivers **204A**, **204B** for the embodiment of FIG. 7, or a driving component of the transmission **208** for the embodiment shown in FIG. 8), and the outer end **221** may be farthest or most distal to the corresponding driving component.

[0061] In addition, each of the wheels **202A**, **202B** includes a radially outer surface **222** that extends between the ends **220**, **221** generally along the axis **225**. The radially outer surface **222** is tapered. Specifically, for the first wheel **202A**, the radially outer surface **222** may generally taper radially inward (or radially toward the axis **225**) when moving axially from the inner end **220** to the outer end **221**, and for the second wheel **202B**, the radially outer surface **222** may generally taper radially outward (or radially away from the axis **225**) when moving axially from the inner end **220** to the outer end **221**. For both wheels **202A**, **202B** a continuous helical groove **224** may be formed in the radially outer surface **222** that extends helically about the axis **225** between ends **220**, **221**.

The grooves **224** of the wheels **202A**, **202B** may receive the corresponding lifting lines **170**, **172** as the lifting lines **170**, **172** are paid onto the corresponding wheels **202A**, **202B** during operations.

[0062] The tapered radially outer surfaces **222** of the wheels **202A**, **202B** may allow the lifting lines **170**, **172** (FIGS. 7 and 8) to pay in and out at different rates relative to the wheels **202A**, **202B** as the wheels **202A**, **202B** are rotated about the corresponding axes **225** at a constant rotational speed. For instance, FIGS. 10(a) and 10(b) illustrate the lifting lines **170**, **172** paid onto the wheels **202A**, **202B** when the motor **152** is at (or proximate to) the positions (I) and (II), respectively, in FIGS. 7 and 8 according to some embodiments.

[0063] As shown in FIG. 10(a), at the first position (I) in FIGS. 7 and 8, and as the wheels **202A**, **202B** rotate about the corresponding axes **225** in the direction **227**, the first lifting line **170** and second lifting line **172** may be paid out/in from the grooves **224** of wheels **202A**, **202B** closer to the inner ends **220**. As previously described, the radially outer surfaces **222** of the wheels **202A**, **202B** are tapered such that an outer diameter of the first wheel **202A** may be larger at the inner end **220** than at the outer end **221**, while an outer diameter of the second wheel **202B** may be smaller at the inner end **220** than at the outer end **221**. As a result, during movement from the first position (I) (FIGS. 7 and 8) to the second position (II), the initial pay out of the first lifting line **170** from the first wheel **202A** may be faster than the initial pay in of the second lifting line **172** on the second wheel **202B**.

[0064] Conversely, as shown in FIG. 10(b), at the second position (II) in FIGS. 7 and 8, and as the wheels **202A**, **202B** continue to rotate about the corresponding axes **225** in the direction **227**, the first lifting line **170** and second lifting line **172** may be paid out from the grooves **224** of wheels **202A**, **202B** closer to the outer ends **221**. As previously described, the radially outer surfaces **222** of the wheels **202A**, **202B** are tapered such that an outer diameter of the first wheel **202A** may be smaller at the outer end **221** than at the inner end **220**, while an outer diameter of the second wheel **202B** may be larger at the outer end **221** than at the inner end **221**. As a result, during movement from the first position (I) (FIGS. 7 and 8) to the second position (II), as the motor **152** approaches the second position (II) the pay out of the first lifting line **170** from the first wheel **202A** may be slower than the pay in of the second lifting line **172** onto the second wheel **202B**. Moreover, the diameters of the wheels **202A**, **202B** may be generally equal at a mid-point or middle portion of the wheels **202A**, **202B** between the corresponding ends **220**, **221**, so that the pay out and pay in of the lifting lines **170** and **172**, respectively, may be generally at the same rate at the mid-point between the positions (I), (II). As previously described, these relative changes in the pay in and pay out rates of the lifting lines **170**, **172** from the wheels **202A**, **202B** may be configured to move the motor **152** laterally between the positions (I) and (II) with out or with minimal vertical deviation. However, by configuring the wheels **202A**, **202B** to have the tapered outer diameter as shown in FIGS. 9 and 10 and described herein, the varied rates of pay for the lifting lines **170**, **172** relative to the wheels **202A**, **202B** may be achieved with a constant rotational speed of the wheels **202A**, **202B** about the

axes **225** (e.g., in the rotational direction **227** shown in FIGS. **10(a)** and **10(b)**).

[0065] Referring now to FIG. **11**, in some embodiments, a single, continuous lifting line **270** may be coupled to each of the pair of wheels **202A**, **202B** and the motor **152** (or more particularly the lifting eye **153**) so that a rotation of the wheels **202A**, **202B** may translate the motor between the position (I), (II), (III). Specifically, in the example embodiment shown in FIG. **11**, the wheels **202A**, **202B** may comprise gears or sprockets, and the continuous lifting line **270** may comprise a chain or geared belt. As a result, rotation of the wheels **202A**, **202B** may advance the lifting line **270** over the wheels **202A**, **202B** to thereby laterally move the motor **152** between the positions (I), (II), (III).

[0066] Specifically, rotating the wheels **202A**, **202B** so as to further circulate the lifting line **270** in a generally “clockwise” rotation as shown in the view of FIG. **11** may cause the motor **152** to advance from the first position (I) to the second position (II) and ultimately to the third position (III). Conversely, rotating the wheels **202A**, **202B** so as to further circulate the lifting line **270** in a generally “counter clockwise” rotation as shown in the view of FIG. **11** may cause the motor **152** to advance from the third position (III) to the second position (II) and ultimately the first position (I).

[0067] Because the rotation of the wheels **202A**, **202B** and the general advance of the lifting line **270** over the wheels **202A**, **202B** is synchronized in the embodiment shown in FIG. **11**, the motor **152** may tend to deviate vertically as it moves laterally between the positions (I), (II), (III).

Specifically, when the motor **152** is at the second position (II), which is midway between the first position (I) and the third position (III) in FIG. **11**, the motor **152** may be vertically lower than when the motor **152** is positioned at either the first position (I) or the third position (III).

[0068] In the example embodiment of FIG. **11**, a single driver (e.g., driver **206**, **204A**, **204B**, etc.) may be used to drive the rotation of one or both of the wheels **202A**, **202B** to traverse the motor **152** laterally between the positions (I), (II), (III). Alternatively, in some embodiments, each of the wheels **202A**, **202B** may be rotated by a different driver (e.g., drivers **204A**, **204B** shown in FIG. **7**) to traverse the motor **152** laterally between the positions (I), (II), (III). Without being limited to this or any other theory, use of a single, continuous lifting line **270** may simplify the control and operation of the system for laterally traversing the motor **152** across the enclosure **110** (FIGS. **1** and **2**) by reducing the number of lifting lines and independent lifting points.

[0069] FIGS. **12** and **13** illustrate methods **300**, **400** of moving a motor within an enclosure of an air handler according to some embodiments disclosed herein. In some embodiments, the methods **300**, **400** may be practiced using the systems and features shown in FIGS. **1-11** and described herein. Thus, continuing reference is made to FIGS. **1-11** when describing the features of methods **300**, **400** in FIGS. **12** and **13**. However, such reference should not be interpreted as limiting the systems that may be used to practice the methods **300**, **400**. As a result, methods **300**, **400** may be practiced using system that differ in at least some respects from those shown in FIGS. **1-11** and/or described herein.

[0070] Referring first to FIG. **12**, initially method **300** includes suspending a motor from a first lifting support and second lifting support via lifting line at block **302**. The first lifting support is laterally spaced from the second lifting support in an enclosure of an air handler of a climate control system. For instance, as shown in FIGS. **3A-3G**, a motor **152'** may be suspended from lifting line **170**, **172** that is connected to a pair of lifting supports **162** that are laterally spaced from one another in the enclosure **110**. The lifting supports may comprise any suitable frame, bracket, eyelet, post, winch wheel (e.g., pulley, sprocket, gear, etc.) that is configured to support a tensioned lifting line (e.g., rope, cable, chain, etc.) for suspending another object (e.g., such as a motor **152**).

[0071] In addition, method **300** includes shortening the lifting line extending between the motor and the second lifting support at block **304**, and lengthening the lifting line extending between the motor and the first lifting support at block **306**. The shortening and lengthening at blocks **304** and **306**, respectively, may be performed at least partially at the same time or may be performed at different times (e.g., sequentially).

[0072] In some embodiments, the lifting line may comprise a first lifting line extending between the motor and the first lifting support (e.g., such as the first lifting line **170** shown in FIGS. **3C** extending between the motor **152'** and the first lifting support **162A**) and a second lifting line extending between the motor and the second lifting support (e.g., such as the second lifting line **172** shown in FIG. **3C** extending between the motor **152'** and the second lifting support **162B**). Thus, shortening the lifting line at block **304** may comprise shortening (e.g., via a winch, wheel, etc.) the second lifting line between the motor and the second lifting support, and lengthening the lifting line at block **306** may comprise extending (e.g., again via winch, wheel, etc.) the first lifting line between the motor and the first lifting support.

[0073] In some embodiments, the lifting line may comprise a continuous lifting line that is connected to each of the first lifting support, the second lifting support, and the motor (e.g., such as is the case for lifting line **270** shown in FIG. **11**). In addition, the first lifting support and the second lifting support may comprise first and second wheels (e.g., pulleys, gears, sprockets, etc.). Thus, shortening the lifting line at block **304** and lengthening the lifting line at block **306** may comprise rotating the first wheel (first lifting support) and the second wheel (second lifting support) to shorten the length of the lifting line extending between the second lifting support and the motor and increasing the length of the lifting line extending between the first lifting support and the motor.

[0074] Referring still to FIG. **12**, method **300** also includes, at block **308**, moving the motor laterally within the enclosure as a result of the shortening and lengthening in blocks **304** and **306**, respectively. For instance, as previously described and shown in FIGS. **3A-3G**, **4A-4C**, **7**, **8**, **11**, lengthening the lifting line (e.g., first lifting line **170**, lifting line **270**) between the motor (e.g., motor **152**) and the first lifting support (e.g., lifting support **162A**) and shortening the lifting line between the motor and the second lifting support (e.g., lifting support **162B**) may laterally traverse the motor between the lifting supports and thus laterally within the enclosure (e.g., enclosure **110**).

[0075] Referring now to FIG. **13**, initially method **400** includes, at block **402**, lifting a motor off of a motor base in an enclosure of an air handler with a first lifting line that is connected to a first lifting support in the enclosure. For instance, as shown in FIGS. **3A** and **4A**, the motor **152'** may initially be lifted off of the corresponding motor base by a first lifting line **170** that is connected to a first lifting support **162A** in the enclosure.

[0076] In addition, method **400** includes connecting the motor to a second lifting support with a second lifting line at block **404**. The second lifting support may be positioned laterally closer to an access door of the enclosure than the first lifting support, such that the first lifting support and the second lifting support may be laterally spaced from one another in the enclosure. For instance, as shown in FIGS. **3B** and **4B**, after the motor **152'** is lifted off of the motor base **151** via the first lifting line **170**, a second lifting line **172** is connected between the motor **152'** and a second lifting support **162B** that is laterally spaced from the first lifting support **162A** so that the second lifting support **162B** is closer to the access door **116**.

[0077] Further, method **400** includes lengthening the first lifting line between the motor and the first lifting support at block **406** and shortening the second lifting line between the motor and the second lifting support at block **408**, and laterally moving the motor toward the access door at block **410**. Blocks **406**, **408**, **410** may be similar to blocks **304**, **306**, **308** in method **300** (FIG. **12**). Thus, in some embodiments, blocks **406**, **408**, **410** may include shortening (e.g., via a winch, wheel, etc.) the second lifting line between the motor and the second lifting support, and extending (e.g., again via winch, wheel, etc.) the first lifting line between the motor and the first lifting support, to thereby laterally move the motor between the first lifting support and the second lifting support in the enclosure.

[0078] As previously described, in some embodiments, the lifting beam **160** may be omitted (or substantially shortened) and one or more of the lifting supports **162** may be directly and individually supported via the structure of the enclosure **110** (e.g., walls **112**, ceiling **114**, floor **115**, etc.). Without being limited to this or any other theory, a continuous lifting beam that spans all or

part of the enclosure **110** may require additional construction and support framing within and about the enclosure **110**. Thus, shortening or avoiding the lifting beam **160** may simplify the construction of the enclosure **110**.

[0079] Embodiments disclosed herein include systems and methods for moving safely and efficiently moving a fan motor within the confined space of an enclosure of an air handler of a climate control system. In some embodiments, the systems and methods described herein may include controlled and coordinated lengthening and shortening of lengths of lifting lines connected to laterally spaced lifting supports within the enclosure so as to laterally traverse and maneuver the fan motor while minimizing uncontrolled movements (e.g., swinging). Thus, through use of the embodiments disclosed herein, a fan motor may be efficiently and safely moved about the limited space of an enclosure of climate control system air handler module with minimal additional infrastructure.

[0080] While specific embodiments described herein are directed to moving a motor within an enclosure of an air handler of a climate control system, it should be appreciated that the embodiments disclosed herein may be used to lift and move other objects from overhead supports. For instance, in some embodiments, the processes and methods described herein may be utilized to lift other components, assemblies, systems, etc. within or about the air handler **100**, such as, for instance, components of one of the fan assemblies **150** (FIG. 1) including the air inlet **158**, impeller **156**, a motor **152** or any combination thereof. In addition, embodiments of the processes and methods described herein may be generally used to move heavy objects laterally between spaced overhead lifting supports with minimal infrastructure, both for other components or systems of a climate control system, and even for applications that are outside of air handlers or climate control systems entirely.

[0081] As explained above and reiterated below, the present disclosure includes, without limitation, the following example implementations.

[0082] Clause 1: A method of moving a motor within an enclosure of an air handler of a climate control system, the method comprising: (a) suspending the motor from a first lifting support and a second lifting support via lifting line, the first lifting support being laterally spaced from the second lifting support in the enclosure; (b) shortening the lifting line extending between the motor and the second lifting support; (c) lengthening the lifting line extending between the motor and the first lifting support; and (d) moving the motor laterally within the enclosure as a result of (b) and (c).

[0083] Clause 2: The method of any of the clauses, wherein the lifting line comprises a continuous lifting line that extends between the motor, the first lifting support, and the second lifting support.

[0084] Clause 3: The method of any of the clauses, wherein the first lifting support comprises a first wheel and the second lifting support comprise a second wheel, wherein (b) and (c) comprise rotating the first wheel and the second wheel.

[0085] Clause 4: The method of any of the clauses, wherein (b) and (c) comprise rotating at least one of the first wheel and the second wheel with a driver.

[0086] Clause 5: The method of any of the clauses, wherein the continuous lifting line comprises a chain, and wherein at least one of the first wheel and the second wheel comprises a sprocket.

[0087] Clause 6: The method of any of the clauses, wherein the lifting line comprises a first lifting line and a second lifting line, and wherein (a) comprises: (a1) connecting the motor to the first lifting support with the first lifting line; and (a2) connecting the motor to the second lifting support with the second lifting line.

[0088] Clause 7: The method of any of the clauses, wherein either (b) or (c) further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a manual winch.

[0089] Clause 8: The method of any of the clauses, wherein the manual winch is selected from the group consisting of a come-along, a chain hoist, or a ratchet.

[0090] Clause 9: The method of any of the clauses, wherein either (b) or (c) further comprises

further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a driver.

[0091] Clause 10: The method of any of the clauses, wherein the first lifting support comprises a first wheel and the second lifting support comprises a second wheel, wherein (b) comprises paying the second lifting line into the second wheel; and wherein (c) comprises paying the first lifting line out from the first wheel.

[0092] Clause 11: The method of any of the clauses, wherein (b) comprises paying the second lifting line into the second wheel at a first rate, and wherein (c) comprise paying the first lifting line out from the first wheel at a second rate, the second rate being different from the first rate.

[0093] Clause 12: The method of any of the clauses, wherein (d) comprises substantially maintaining the motor in a lateral plane while moving the motor laterally within the enclosure.

[0094] Clause 13: A method of removing a motor from an enclosure of an air handler of a climate control system, the method comprising: (a) lifting the motor off of a motor base in the enclosure with a first lifting line that is connected to a first lifting support in the enclosure; (b) connecting the motor to a second lifting support with a second lifting line after (a), wherein the second lifting support is laterally closer to an access door of the enclosure than the first lifting support; (c) lengthening the first lifting line between the motor and the first lifting support; (d) shortening the second lifting line between the motor and the second lifting support; and (e) laterally moving the motor toward the access door as a result of (c) and (d).

[0095] Clause 14: The method of any of the clauses, wherein either (b) or (c) further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a manual winch.

[0096] Clause 15: The method of any of the clauses, wherein the manual winch is selected from the group consisting of a come-along, a chain hoist, or a ratchet.

[0097] Clause 16: The method of any of the clauses, wherein (d) comprises shortening the second lifting line between the motor and the second lifting support during (c).

[0098] Clause 17: The method of any of the clauses, further comprising substantially maintaining the motor in a lateral plane during (e).

[0099] Clause 18: The method of any of the clauses, wherein (c) comprises lengthening the first lifting line between the motor and the first lifting support at a first rate, wherein (d) comprises shortening the second lifting line between the motor and the second lifting support at a second rate, wherein the first rate and the second rate are different.

[0100] Clause 19: The method of any of the clauses, wherein the first lifting support comprises a first wheel and the second lifting support comprises a second wheel, wherein (c) comprises rotating the first wheel at a first rotational speed, and wherein (d) comprises rotating the second wheel at a second rotational speed, wherein the first rotational speed and the second rotational speed are equal.

[0101] Clause 20: The method of any of the clauses, wherein the first wheel and the second wheel each comprise a tapered wheel having a continuous helical groove formed thereon.

[0102] The preceding discussion is directed to various exemplary embodiments. However, one of ordinary skill in the art will understand that the examples disclosed herein have broad application, and that the discussion of any embodiment is meant only to be exemplary of that embodiment, and not intended to suggest that the scope of the disclosure, including the claims, is limited to that embodiment.

[0103] The drawing figures are not necessarily to scale. Certain features and components herein may be shown exaggerated in scale or in somewhat schematic form and some details of conventional elements may not be shown in interest of clarity and conciseness.

[0104] In the discussion herein and in the claims, the terms “including” and “comprising” are used in an open-ended fashion, and thus should be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” is intended to mean either an indirect or direct connection.

Thus, if a first device couples to a second device, that connection may be through a direct connection of the two devices, or through an indirect connection that is established via other devices, components, nodes, and connections. In addition, as used herein, the terms “axial” and “axially” generally mean along or parallel to a given axis (e.g., central axis of a body or a port), while the terms “radial” and “radially” generally mean perpendicular to the given axis. For instance, an axial distance refers to a distance measured along or parallel to the axis, and a radial distance means a distance measured perpendicular to the axis. Further, when used herein (including in the claims), the words “about,” “generally,” “substantially,” “approximately,” and the like, when used in reference to a stated value mean within a range of plus or minus 10% of the stated value. [0105] While exemplary embodiments have been shown and described, modifications thereof can be made by one skilled in the art without departing from the scope or teachings herein. The embodiments described herein are exemplary only and are not limiting. Many variations and modifications of the systems, apparatus, and processes described herein are possible and are within the scope of the disclosure. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. Unless expressly stated otherwise, the steps in a method claim may be performed in any order. The recitation of identifiers such as (a), (b), (c) or (1), (2), (3) before steps in a method claim are not intended to and do not specify a particular order to the steps, but rather are used to simplify subsequent reference to such steps.

Claims

1. A method of moving a motor within an enclosure of an air handler of a climate control system, the method comprising: (a) suspending the motor from a first lifting support and a second lifting support via lifting line, the first lifting support being laterally spaced from the second lifting support in the enclosure; (b) shortening the lifting line extending between the motor and the second lifting support; (c) lengthening the lifting line extending between the motor and the first lifting support; and (d) moving the motor laterally within the enclosure as a result of (b) and (c).
2. The method of claim 1, wherein the lifting line comprises a continuous lifting line that extends between the motor, the first lifting support, and the second lifting support.
3. The method of claim 2, wherein the first lifting support comprises a first wheel and the second lifting support comprise a second wheel, wherein (b) and (c) comprise rotating the first wheel and the second wheel.
4. The method of claim 3, wherein (b) and (c) comprise rotating at least one of the first wheel and the second wheel with a driver.
5. The method of claim 4, wherein the continuous lifting line comprises a chain, and wherein at least one of the first wheel and the second wheel comprises a sprocket.
6. The method of claim 1, wherein the lifting line comprises a first lifting line and a second lifting line, and wherein (a) comprises: (a1) connecting the motor to the first lifting support with the first lifting line; and (a2) connecting the motor to the second lifting support with the second lifting line.
7. The method of claim 6, wherein either (b) or (c) further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a manual winch.
8. The method of claim 7, wherein the manual winch is selected from the group consisting of a come-along, a chain hoist, or a ratchet.
9. The method of claim 6, wherein either (b) or (c) further comprises further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a driver.
10. The method of claim 6, wherein the first lifting support comprises a first wheel and the second lifting support comprises a second wheel, wherein (b) comprises paying the second lifting line into the second wheel; and wherein (c) comprises paying the first lifting line out from the first wheel.
11. The method of claim 10, wherein (b) comprises paying the second lifting line into the second

wheel at a first rate, and wherein (c) comprise paying the first lifting line out from the first wheel at a second rate, the second rate being different from the first rate.

12. The method of claim 11, wherein (d) comprises substantially maintaining the motor in a lateral plane while moving the motor laterally within the enclosure.

13. A method of removing a motor from an enclosure of an air handler of a climate control system, the method comprising: (a) lifting the motor off of a motor base in the enclosure with a first lifting line that is connected to a first lifting support in the enclosure; (b) connecting the motor to a second lifting support with a second lifting line after (a), wherein the second lifting support is laterally closer to an access door of the enclosure than the first lifting support; (c) lengthening the first lifting line between the motor and the first lifting support; (d) shortening the second lifting line between the motor and the second lifting support; and (e) laterally moving the motor toward the access door as a result of (c) and (d).

14. The method of claim 13, wherein either (b) or (c) further comprises changing a length of the first lifting line or the second lifting line, respectively, by use of a manual winch.

15. The method of claim 14, wherein the manual winch is selected from the group consisting of a come-along, a chain hoist, or a ratchet.

16. The method of claim 13, wherein (d) comprises shortening the second lifting line between the motor and the second lifting support during (c).

17. The method of claim 16, further comprising substantially maintaining the motor in a lateral plane during (e).

18. The method of claim 16, wherein (c) comprises lengthening the first lifting line between the motor and the first lifting support at a first rate, wherein (d) comprises shortening the second lifting line between the motor and the second lifting support at a second rate, wherein the first rate and the second rate are different.

19. The method of claim 18, wherein the first lifting support comprises a first wheel and the second lifting support comprises a second wheel, wherein (c) comprises rotating the first wheel at a first rotational speed, and wherein (d) comprises rotating the second wheel at a second rotational speed, wherein the first rotational speed and the second rotational speed are equal.

20. The method of claim 19, wherein the first wheel and the second wheel each comprise a tapered wheel having a continuous helical groove formed thereon.
